C.W
20107

BSc D.S
Pritee Maam
ICS

Number Systems and Conversion

- Decimal → base : 10 (0-9)
- Binary → base : 2 (0,1)
- Octal → base : 8 (0-7)
- Hexadecimal → base : 16 (0-9, A-F)

Conversion: B, O, H → Multiplication → $(D)_{10}$ → Division → B, O, H

$B = (101011)_2 = (1\times2^5 + 0\times2^4 + 1\times2^3 + 0\times2^2 + 1\times2^1 + 1\times2^0)$

$= (43)_{10}$

$O = (723)_8 = 7\times8^2 + 2\times8^1 + 3\times8^0 = (467)_{10}$

$H = (5F9)_{16} = 5\times16^2 + 15\times16^1 + 9\times16^0 = (1529)_{10}$

$B = (1010)_2 = (1\times2^3 + 0\times2^2 + 1\times2^1 + 0\times2^0) = (10)_{10}$

$O = (45)_8 = (4\times8^1 + 5\times8^0) = (37)_{10}$

$H = (EA)_{16} \rightarrow (14\times16^1 + 10\times16^0) = (234)_{10}$

$D = (234)_{10} =$

2 | 234	
2 | 117 - 0	2 | 7 - 0
2 | 58 - 1	2 | 3 - 1
2 | 29 - 0	2 | 1 - 1
2 | 14 - 1	0 - 1
2 | 7 - 0	

$(11101010)_2$

$D = (234)_{80} =$ 8 | 234 ; 8 | 29 – 2 ; 8 | 3 – 5 ; 0 – 3 $= (352)_8$

$D = (234)_{10} =$ 16 | 234 ; 16 | 14 – 10 – A ; 0 – 14 – E $= (EA)_{16}$

$D = (658)_{10} =$ 8 | 658 ; 8 | 82 – 2 ; 8 | 10 – 2 ; 8 | 1 – 2 ; 0 – 1 $= (1222)_8$

$D = (658)_{10} =$ 2 | 658 ; 2 | 329 – 0 ; 2 | 164 – 1 ; 2 | 82 – 0 ; 41 – 0

2 | 41 – 0 ; 2 | 20 – 1 ; 2 | 10 – 0 ; 2 | 5 – 0 ; 2 | 2 – 1 ; 0 – 1 $= (1010010010)_2$

$D = (658)_{10} =$ 16 | 658 ; 16 | 41 – 2 ; 16 | 2 – 9 ; 0 – 2 $= (292)_{16}$

$(1100011)_2 \longrightarrow (1\times2^6+1\times2^5+0\times2^4+0\times2^3+0\times2^2+1\times2^1+1\times2^0) = (99)_{10}$

$(99)_{10} \rightarrow$ 8 | 99 ; 8 | 12 – 3 ; 8 | 1 – 4 ; 0 – 1 $= (143)_8$

$(456)_8 \rightarrow 4\times8^2+5\times8^1+6\times8^0 = (302)_{10} =$ ~~$3\times2^2+0\times2$~~

16 | 302 ; 16 | 18 – E ; 16 | 1 – 2 ; 0 – 1 $= (12E)_{16}$

$(2AC)_{16} = 2\times16^2 + 10\times16^1 + 12\times16^0 = (684)_{10}$

$$
\begin{array}{r|l}
2 & 684 \\
\hline
2 & 342 - 0 \\
\hline
2 & 171 - 0 \\
\hline
 & 85 - 1
\end{array}
\qquad
\begin{array}{r|l}
2 & 42 - 1 \\
\hline
2 & 21 - 0 \\
\hline
2 & 10 - 1 \\
\hline
 & 5 - 0
\end{array}
\qquad
\begin{array}{r|l}
2 & 2 - 1 \\
\hline
2 & 1 - 0 \\
\hline
 & 0 - 1
\end{array}
$$

$= (1010101100)_2$